

MOSAIC: Multi-Omics Subtype Analysis In Cancer

Latent Transformation Trajectories in Lung Adenocarcinoma

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Abstract

Lung adenocarcinoma (LUAD) exhibits substantial inter-patient molecular heterogeneity that transcriptomic subtyping of tumour tissue only partially captures. Here we ask whether this heterogeneity extends to the tumourigenesis *process* itself: do patients undergo qualitatively distinct molecular transformations? We apply Multi-Omics Factor Analysis (MOFA+) to 102 matched tumour-normal pairs from the Clinical Proteomic Tumour Analysis Consortium LUAD cohort, learning a 13-factor latent space from transcriptomic, proteomic, and phosphoproteomic data. For each patient we compute a latent transformation vector $\Delta Z = Z_{\text{tumour}} - Z_{\text{normal}}$, encoding the direction and magnitude of molecular change in the shared latent space. Clustering these vectors by k-means identified three transformation subtypes that are reproducible across alternative representations (ARI ≥ 0.80). One-vs-rest differential expression followed by preranked GSEA revealed systematic differences in transcriptional reprogramming: subtype 0 shows relatively suppressed proliferative reprogramming and is significantly enriched for *EGFR* mutations; subtype 1 is dominated by E2F and G2-M cell-cycle activation; and subtype 2 combines interferon signalling with a proliferative component. These signatures are broadly consistent with the TRU, PP, and PI state-based LUAD subtypes respectively, though the correspondence is suggestive rather than formally established. No significant overall survival difference was detected across subtypes ($n = 102$). The analysis provides a proof-of-concept that latent transformation vectors derived from a matched multi-omics cohort can reveal biologically coherent modes of tumourigenesis.

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1 Introduction

Lung adenocarcinoma (LUAD) is the most common subtype of non-small cell lung cancer and exhibits substantial molecular heterogeneity. Transcriptomic profiling of tumour tissue by the Cancer Genome Atlas (TCGA) identified three reproducible molecular subtypes: the terminal respiratory unit (TRU), characterised by differentiation markers and enrichment for *EGFR* mutations; the proximal-inflammatory (PI) subtype, dominated by immune-related gene expression; and the proximal-proliferative (PP) subtype, marked by cell cycle and *KRAS*-associated programmes [1]. These subtypes describe the tumour *state* — the molecular profile of established disease. Matched tumour-normal cohorts, such as the *Clinical Proteomic Tumour Analysis Consortium LUAD dataset* [2], were designed to support a complementary analysis: by profiling adjacent normal lung tissue from the same patient, they make it possible to attribute molecular differences to the tumour itself rather than to inter-individual variation in normal tissue composition. In practice, matched designs are most often used for conventional paired differential expression, which identifies genes that shift on average across the cohort but discards patient-level information about how far and in which direction any individual tumour has moved.

The central question of this project is whether patients with LUAD undergo qualitatively distinct molecular transformations during tumourigenesis, or whether transformation is essentially uniform across the cohort. We address this by operating in a multi-omics latent space rather than in the original high-dimensional feature space. Multi-Omics Factor Analysis (MOFA+) [3] is fit jointly on all tumour and normal samples, yielding a set of latent factors that capture shared axes of variation across transcriptomic, proteomic, and phosphoproteomic data. Every sample receives a factor score vector Z ; for each patient i we define the *latent transformation vector*

$$\Delta Z_i = Z_{\text{tumour},i} - Z_{\text{normal},i}, \quad (1)$$

which encodes the direction and magnitude of that patient’s tumourigenesis as a displacement in the shared latent space. Clustering patients by ΔZ thus groups those who undergo qualitatively similar multi-omics reprogramming, regardless of their absolute tumour state. We refer to these groups as *transformation subtypes* to distinguish them from the state-based subtypes identified by prior transcriptomic classification.

This report describes the full MOSAIC pipeline. We first fit a MOFA+ model on all 204 samples to learn a compact, biologically grounded latent representation (§2). We then compute per-patient ΔZ vectors and characterise their geometry — magnitude, factor composition, and relationships to clinical covariates. We cluster the ΔZ vectors to identify candidate transformation subtypes and assess their robustness across alternative representations. We interpret each subtype biologically using pathway analysis on both MOFA factor loadings and per-patient differential expression, and finally test preliminary

clinical associations through survival and driver mutation analyses.

2 Methods

2.1 Dataset and preprocessing

We used the CPTAC-LUAD cohort described by Gillette et al. [2], accessed via the `cptac` Python package. Of the 111 patients in the dataset, nine were excluded because they lacked a complete matched tumour-normal pair in at least one of the three omics modalities, leaving 102 patients (204 samples) for analysis. The three modalities were RNA-seq (BCM pipeline, reported as TPM and converted to $\log_2(\text{TPM} + 1)$), proteomics (Michigan CPTAC pipeline, log-ratio), and phosphoproteomics (Michigan, log-ratio). Missing values were handled as follows: features absent in more than 20% of samples were removed; the remainder were median-imputed column-wise.

Feature selection was applied independently per modality using the median absolute deviation (MAD) of each feature across all 204 samples. The RNA-seq data contained 59,286 features, reflecting the comprehensive GENCODE annotation used by the BCM pipeline and including protein-coding genes, long non-coding RNAs, and pseudogenes. We retained the top 5,000 transcripts, 5,000 proteins, and 6,000 phosphosites by MAD. The retained features were then z-scored per feature using a single scaler fit jointly on all 204 samples; tumour and normal samples were not z-scored separately. This joint scaling is required for the ΔZ framework: both tissue states must share a common reference frame for their difference to be interpretable as a biological transformation.

2.2 Latent factor modelling with MOFA+

We applied Multi-Omics Factor Analysis (MOFA+, [3]) to all 204 samples treated as a single group with three views. MOFA+ decomposes the observed data matrix as a product of per-sample latent factor scores and per-feature loading weights, assuming that each observed value arises from a Gaussian distribution centred on the linear combination of factor contributions.

We initialised 20 factors and fit five models from different random seeds (convergence mode: fast, maximum 1,000 iterations); the model with the highest final evidence lower bound (ELBO) was selected for downstream analysis.

Inactive factors were removed post-hoc: a factor was retained only if the fraction of variance it explained in at least one view exceeded a threshold of $R_{\max}^2 \geq 0.01$. This retained 13 of the initial 20 factors.

2.3 Transformation vectors and clustering

For each of the 102 patients we computed a latent transformation vector as defined in equation (1), using the 13-dimensional factor score vector of each sample. This produces

a 102×13 matrix whose rows encode the direction and magnitude of each patient’s tumourigenesis in the shared MOFA latent space.

In matched tumour-normal data, the first MOFA factor typically captures the dominant axis of variation, which is the tissue-type contrast itself. Variation along this axis reflects the overall extent of transcriptional reprogramming and is shared across patients; it therefore carries less information about patient-specific transformation patterns than the remaining factors. We constructed four representations of the ΔZ matrix to assess sensitivity to this: raw ΔZ using all 13 factors; raw ΔZ with Factor 1 excluded; L2-normalised ΔZ (all factors); and L2-normalised ΔZ with Factor 1 excluded. L2 normalisation places all patients on a unit hypersphere, making clustering respond to transformation direction rather than magnitude. The primary representation for clustering was L2-normalised ΔZ on all 13 factors.

We clustered the 102 patients using k -means (`sklearn.cluster.KMeans`, `n_init=100`), sweeping $k \in \{2, 3, 4, 5, 6\}$ and recording the mean silhouette score at each k . Gaussian Mixture Model (GMM) AIC and BIC were computed at each k as additional descriptive metrics. We selected $k = 3$ based on the silhouette profile, cluster balance (37, 25, and 40 patients), and partition robustness across representations. Robustness was quantified using the adjusted Rand index (ARI), which measures agreement between two independent cluster assignments on a scale from 0 (chance) to 1 (identical). Re-fitting the same algorithm on the three alternative representations and comparing with the primary partition yielded $\text{ARI} = 0.82$ against L2-normalised ΔZ without Factor 1, and $\text{ARI} = 0.80$ against raw ΔZ , indicating that the three-subtype structure is stable across representational choices. Silhouette scores in the range 0.18–0.19 are lower than those typical of well-separated synthetic clusters; however, values in this range are biologically plausible for continuous molecular phenotypes and the decision to proceed with $k = 3$ was supported by the consistency of the partition across representations.

2.4 Biological and clinical interpretation

Factor interpretation. MOFA factor loadings measure the contribution of each gene to a given factor: genes with large positive loadings are the primary drivers of high factor scores, and genes with large negative loadings drive low scores. To characterise the biological programmes encoded in each factor, we ranked all 5,000 transcripts by their RNA loading weight and applied preranked gene set enrichment analysis (GSEA). Preranked GSEA walks down a sorted gene list and tests whether the members of each predefined gene set are concentrated near the top or bottom relative to a permutation null; the normalised enrichment score (NES) indicates the direction and strength of enrichment. A positive NES indicates enrichment among genes with large positive loadings; a negative NES indicates enrichment at the negative-loading pole. We used the MSigDB Hallmark

2020 and KEGG 2021 Human libraries (1,000 permutations, gene set size 15–500) and filtered results at $\text{FDR} < 0.10$ (Benjamini-Hochberg).

Cluster interpretation. To test whether the transformation subtypes differ in their transcriptional reprogramming, we computed per-patient differential expression as $\Delta\text{Expr}_i = \text{RNA}_{\text{tumour},i} - \text{RNA}_{\text{normal},i}$ across the full 59,286-gene matrix from ingestion rather than the MAD-filtered MOFA input. This avoids biasing the test towards features that were pre-selected for latent factor construction; the cluster interpretation is thereby an independent assessment. For each subtype we performed one-vs-rest Mann-Whitney U tests on ΔExpr across all genes and applied Benjamini-Hochberg correction. The per-gene median difference between the focal cluster and all remaining patients was used as the ranking statistic for a second round of preranked GSEA with the same libraries and FDR threshold.

Clinical associations. Overall survival was estimated using the Kaplan-Meier method for each transformation subtype. An omnibus log-rank test was applied to assess whether survival distributions differed across the three subtypes simultaneously. Pairwise log-rank comparisons were conducted only when the omnibus test yielded $p < 0.05$, with Benjamini-Hochberg correction applied. Enrichment of driver mutations (KRAS, EGFR, TP53, ALK) within each subtype was evaluated by two-sided Fisher’s exact test, corrected across all mutation-subtype pairs by Benjamini-Hochberg. Given $n = 102$ patients, all survival analyses should be interpreted as hypothesis-generating; the small cohort size makes it this analysis severely underpowered. Full configuration parameters are available in the project repository at `config/config.yml`.

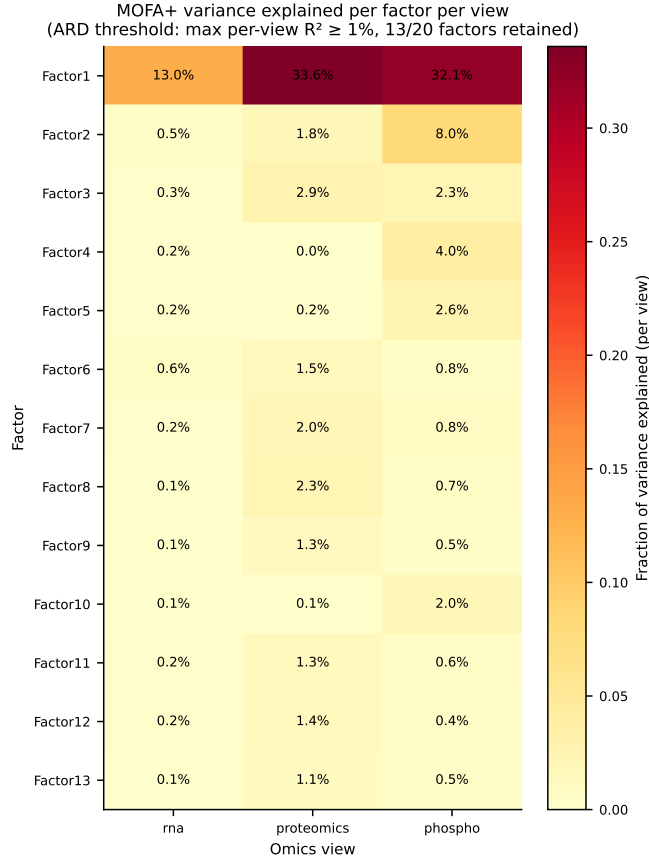


Figure 1: Fraction of variance explained by each retained MOFA+ factor per omics view. Factor 1 explains substantially more variance than any other factor across all three modalities.

3 Results

3.1 MOFA captures a shared multi-omics latent space

Of the 20 initialised factors, 13 were retained after ARD filtering ($R^2_{\max} \geq 0.01$). Factor 1 is by far the dominant component, explaining 13.0% of transcriptomic, 33.6% of proteomic, and 32.1% of phosphoproteomic variance (Figure 1). The remaining 12 factors each explain less than 8.1% of variance in any single view, with most falling below 3%, indicating that variance is heavily concentrated in Factor 1.

Factor 1 perfectly separates tumour from normal tissue: a paired Wilcoxon test on ΔZ yielded $p = 1.8 \times 10^{-18}$ with rank-biserial $r = 1.0$, meaning all 102 patients shifted in the same direction along Factor 1 during tumourigenesis. No other factor reached statistical significance. The PCA trajectory plot in Figure 2 makes this structure visible: PC1 is dominated by Factor 1, and the 102 per-patient arrows are predominantly parallel and oriented along this axis. The consistency of the Factor 1 trajectory confirms that MOFA has learned a biologically coherent shared space in which tumourigenesis appears as a reproducible displacement.

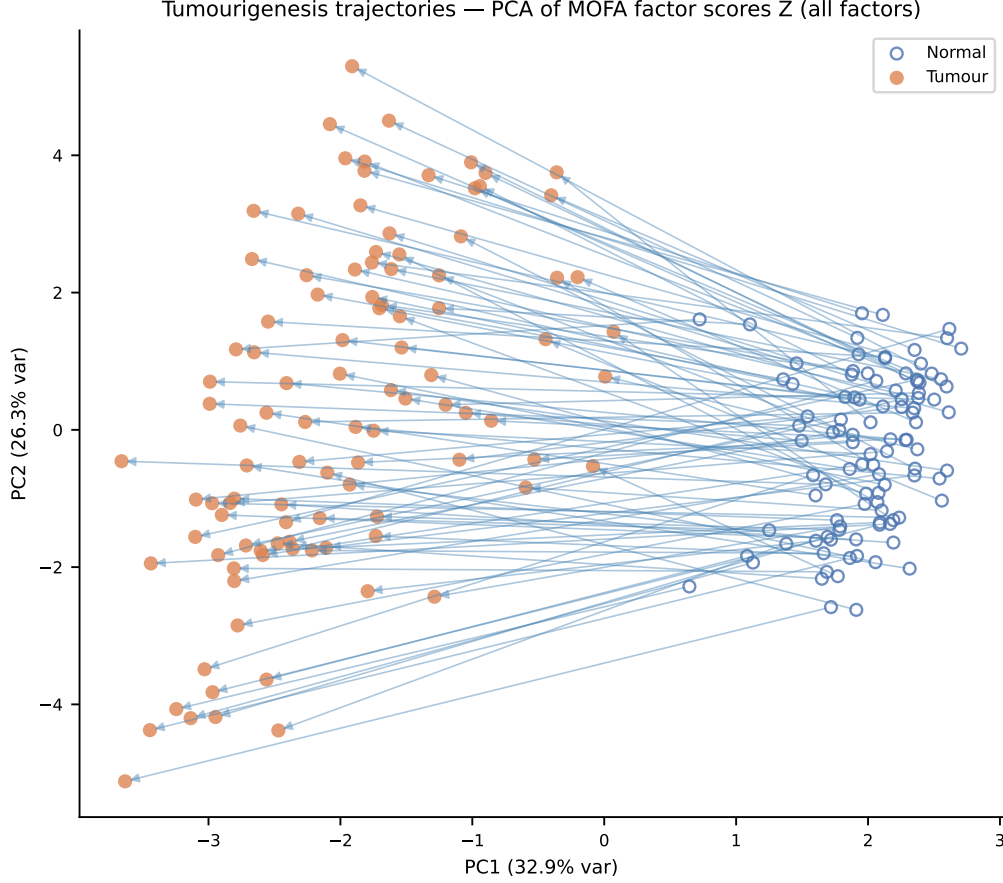


Figure 2: Per-patient tumourigenesis trajectories in the PCA of MOFA+ factor scores (all 204 samples). Each arrow connects the normal (open circle) to the tumour (filled circle) sample of one patient. The dominant direction along PC1 reflects the near-universal Factor 1 shift.

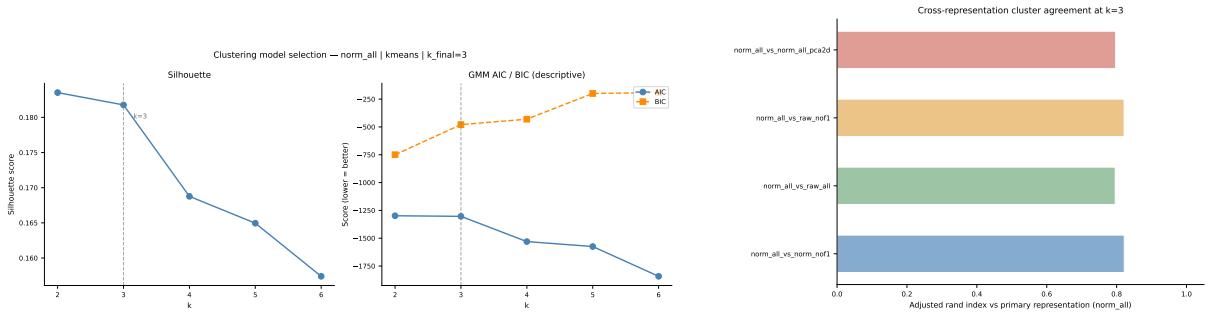
3.2 ΔZ vectors encode patient-level transformation heterogeneity

The ΔZ matrix for 102 patients shows that Factor 1 drives a large, uniform component of tumourigenesis: the rank-biserial correlation of $r = 1.0$ implies that the tumour-minus-normal displacement is in the same direction for every patient along this axis. In contrast, the remaining 12 factors show substantial variation across patients (per-factor distributions are shown in Appendix Figure A7), suggesting that inter-patient heterogeneity in transformation is concentrated in the dimensions beyond the dominant tumour-normal axis.

Transformation magnitude (L2 norm of ΔZ) varied across patients but showed no significant association with tumour stage (Kruskal-Wallis $H = 3.40$, $p = 0.183$; Appendix Figure A8). The absence of a stage-magnitude relationship indicates that transformation extent in latent space does not simply reflect disease progression as captured by clinical staging.

Table 1: Clustering quality metrics across k on L2-normalised ΔZ . Silhouette scores are means over all patients; GMM AIC and BIC are computed independently for each k as descriptive metrics.

k	Silhouette	GMM AIC	GMM BIC
2	0.184	−1298.2	−749.6
3	0.182	−1303.5	−479.3
4	0.169	−1530.2	−430.3
5	0.165	−1574.7	−199.2
6	0.157	−1842.4	−191.3



(a) Silhouette score (left) and GMM AIC/BIC (right) versus k . The silhouette plateau at $k = 2-3$ and the increasing BIC are consistent with weak cluster separation on a continuous manifold.

(b) Adjusted Rand index between the primary partition ($k = 3$, L2-normalised ΔZ) and three alternative representations. All ARI values exceed 0.79.

Figure 3: Clustering model selection (a) and cross-representation robustness (b).

3.3 Clustering identifies three reproducible transformation subtypes

We swept $k \in \{2, 3, 4, 5, 6\}$ on L2-normalised ΔZ . The silhouette score peaked at $k = 2$ (0.184) and remained nearly identical at $k = 3$ (0.182), after which it declined monotonically (Table 1). GMM BIC increased continuously with k , while AIC continued to decrease, indicating that the penalty for additional components outweighs the likelihood gain and that the data do not support strongly separated Gaussian clusters (Figure 3a). We selected $k = 3$ over $k = 2$ based on the nearly indistinguishable silhouette scores, the more balanced partition (37, 25, and 40 patients versus 77 and 25 for $k = 2$), and the fact that $k = 3$ yields a biologically richer description of inter-patient heterogeneity (see §3.4).

The three-subtype partition was robust across representational choices: ARI against L2-normalised ΔZ without Factor 1 was 0.82, against raw ΔZ was 0.80, and against 2D PCA of the primary representation was 0.80 (Figure 3b). Pairwise centroid distances in L2-normalised space ranged from 0.56 (subtypes 0 and 2) to 0.83 (subtypes 0 and 1), exceeding the mean within-cluster radius of 0.48–0.54 for all pairs.

The UMAP projection in Figure 4 and the ΔZ heatmap in Figure 5 illustrate the

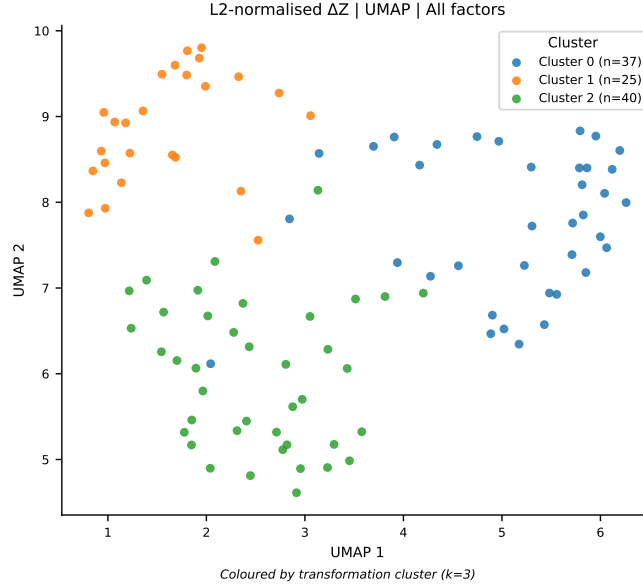


Figure 4: UMAP of L2-normalised ΔZ coloured by transformation subtype. Clustering was performed in the full 13-dimensional space; the UMAP is for visualisation only. Partial overlap between subtypes is consistent with the low silhouette scores.

structure of the three subtypes. In the UMAP, the subtypes occupy partially overlapping regions consistent with a continuous manifold rather than discrete blobs, in agreement with the silhouette scores. The heatmap confirms that all subtypes share a strongly negative Factor 1 component (universal tumourigenesis shift) but differ in their profiles across Factors 2–13, motivating the biological characterisation below.

3.4 Transformation subtypes reflect distinct molecular programmes

Factor interpretation. Preranked GSEA on RNA factor loadings yielded significant results ($\text{FDR} < 0.10$) for 8 of the 13 retained factors; Factors 1, 2, 4, 6, and 9 returned no significant enrichment at this threshold (Figure 6). The most consistently enriched categories were cytokine-cytokine receptor interaction (Factors 3, 5, 13), xenobiotic metabolism and chemical carcinogenesis (Factors 10 and 11), and downstream KRAS signalling suppression (Factors 7 and 12). Factor 1, despite its strong tissue-separation signal, showed no significant pathway enrichment, suggesting that it captures a broad transcriptional state shift that is not well-represented by the curated gene sets used here.

Cluster interpretation. One-vs-rest $\Delta\text{Expression}$ GSEA revealed that all three subtypes are transcriptionally distinct (Figure 7). Subtype 1 ($n = 25$) showed the strongest activation of proliferative programmes: E2F targets ($\text{NES} = 1.80$, $\text{FDR} < 0.001$) and G2-M checkpoint ($\text{NES} = 1.75$, $\text{FDR} = 0.002$) were among the most significantly enriched pathways, indicating that these patients undergo substantially greater tumour-vs-normal upregulation of cell cycle machinery than patients in the other two subtypes. Subtype 0

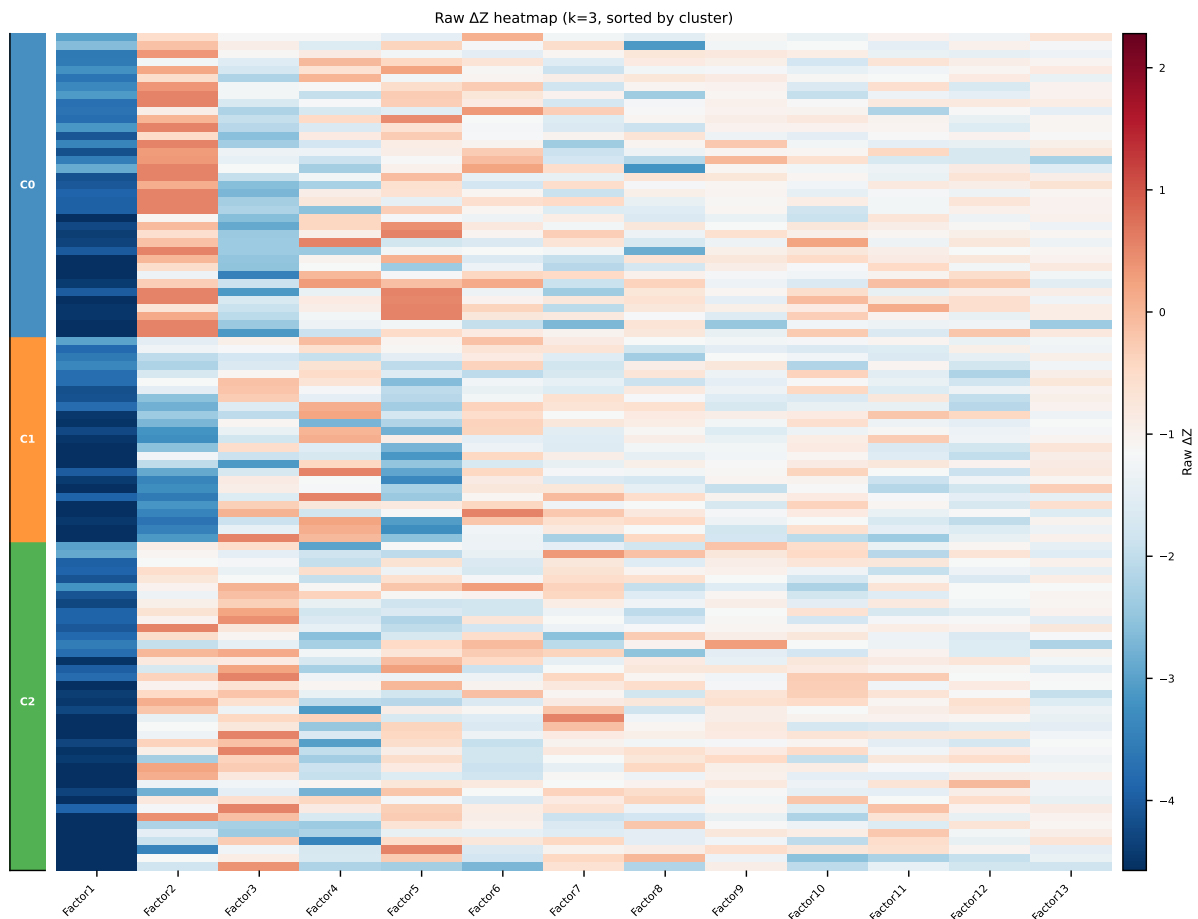


Figure 5: Heatmap of raw ΔZ values (patients $\times$ factors) sorted by transformation subtype. All subtypes share a uniformly negative Factor 1 component; inter-subtype differences are concentrated in Factors 2–13.

($n = 37$) displayed the inverse pattern: the same E2F, G2-M, and MYC target gene sets were negatively enriched ($\text{NES} \leq -1.61$, $\text{FDR} \leq 0.04$), reflecting relatively suppressed proliferative transformation compared to subtypes 1 and 2. Subtype 2 ($n = 40$) showed the strongest interferon response: interferon- α ($\text{NES} = 1.79$, $\text{FDR} = 0.008$) and interferon- γ ($\text{NES} = 1.77$, $\text{FDR} = 0.007$) were the top enriched pathways. E2F targets were also enriched in subtype 2 ($\text{NES} = 1.68$, $\text{FDR} = 0.035$), in contrast to their depletion in subtype 0. The three subtypes therefore differ systematically in their transcriptional reprogramming: subtype 1 is proliferation-dominant, subtype 0 shows relatively suppressed cell-cycle activation, and subtype 2 combines immune activation with a proliferative component.

3.5 Preliminary clinical associations

No significant difference in overall survival was observed across the three subtypes (omnibus log-rank $p = 0.632$; Figure 8). Hence, pairwise comparisons were not performed. With $n = 102$ patients, the study is insufficiently powered to detect modest survival differences.

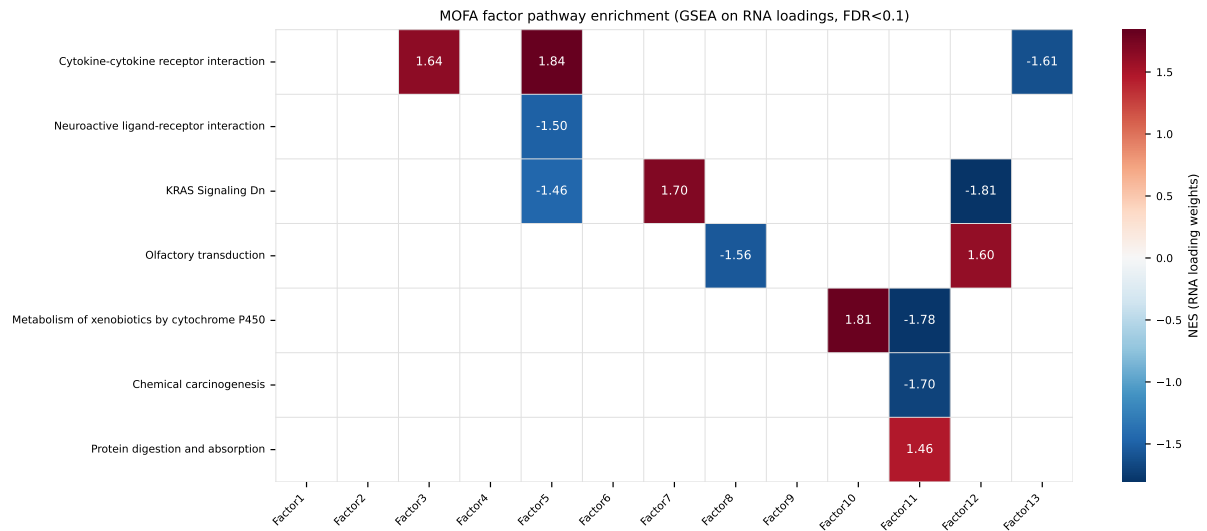


Figure 6: Preranked GSEA results for retained MOFA+ factors using RNA loading weights. Cells show NES; only pathways significant in at least one factor (FDR < 0.10) are displayed. Positive NES indicates enrichment at the positive loading pole; negative NES at the negative pole.

Driver mutation enrichment analysis revealed a significant association between subtype and *EGFR* mutation status (Figure 9). *EGFR* mutations were enriched in subtype 0 (odds ratio 3.18, BH-adjusted $p = 0.043$) and depleted in subtype 1 (odds ratio 0.22, BH-adjusted $p = 0.043$). ALK fusions were strongly depleted in subtype 0 (odds ratio 0.12, BH-adjusted $p = 0.016$), consistent with the mutual exclusivity of *EGFR* mutations and ALK rearrangements as driver events. *KRAS* and *TP53* mutations showed no significant cluster association after Benjamini-Hochberg correction. The co-occurrence of EGFR enrichment, suppressed proliferative transformation, and relatively lower cell-cycle upregulation in subtype 0 is consistent with the biology of EGFR-driven LUAD, which tends towards a more differentiated, TRU-like transcriptional state.

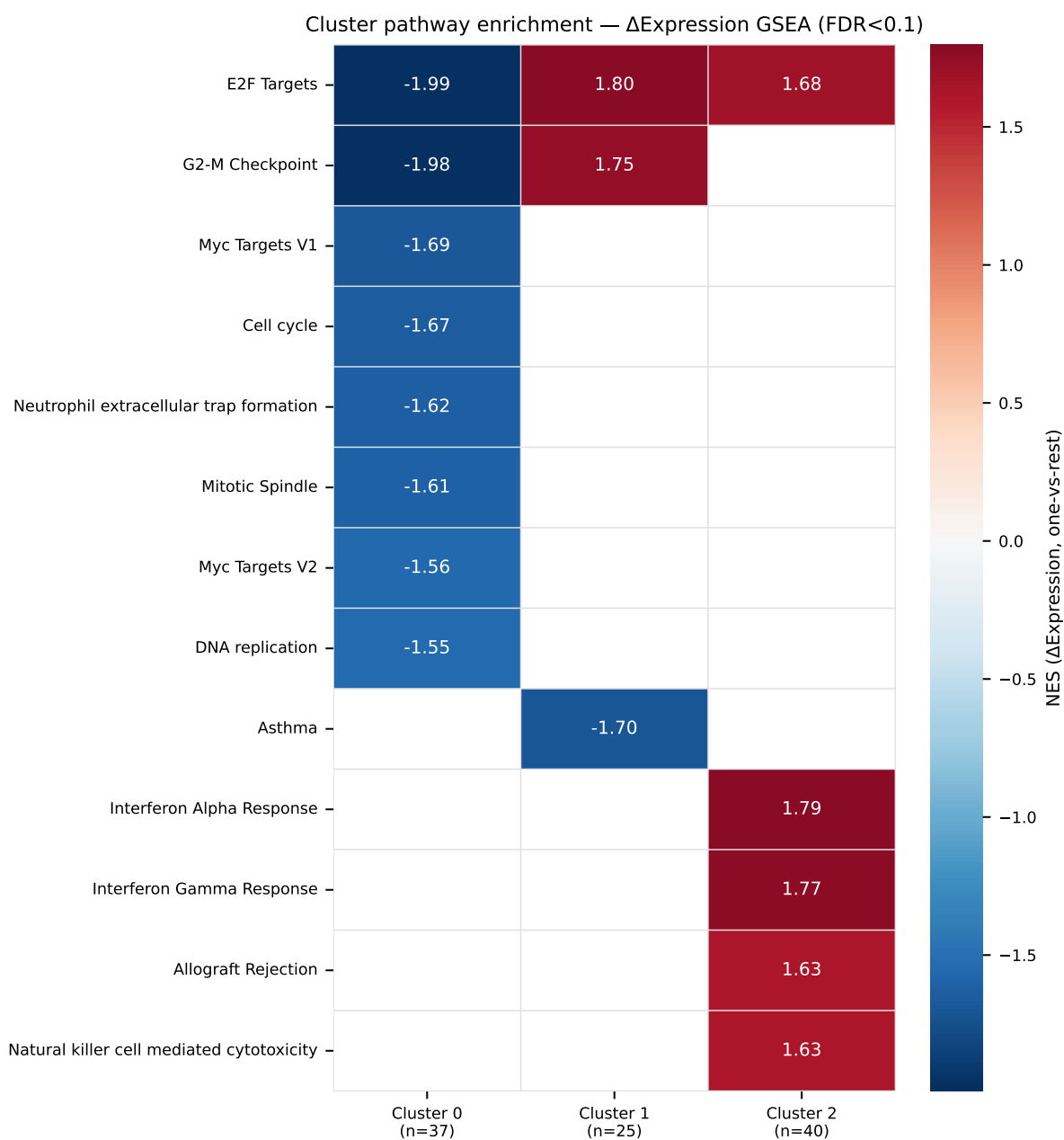


Figure 7: One-vs-rest Δ Expression GSEA results for the three transformation subtypes. Cells show NES; only pathways significant in at least one subtype (FDR < 0.10) are displayed. Negative NES indicates that the pathway is relatively suppressed in the focal subtype compared with the other patients.

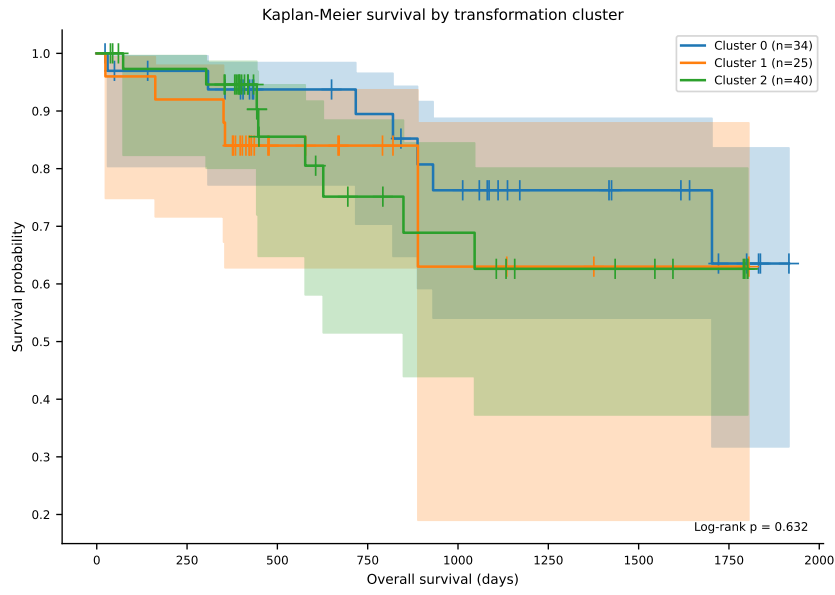


Figure 8: Kaplan-Meier overall survival curves for the three transformation subtypes. Shading shows 95% confidence intervals; tick marks indicate censored observations. Omnibus log-rank $p = 0.632$.

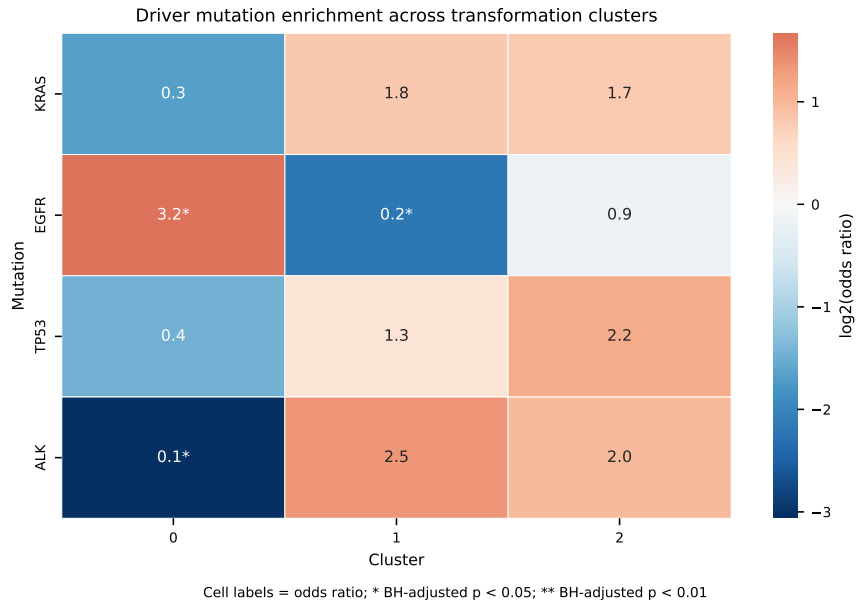


Figure 9: Driver mutation enrichment across transformation subtypes. Cell colour encodes $\log_2$ odds ratio; asterisks indicate BH-adjusted $p < 0.05$ (*) or $p < 0.01$ (**).

4 Discussion

Transformation subtypes in the context of established LUAD biology. The transcriptional signatures of the three subtypes are broadly consistent with the state-based LUAD classification from TCGA [1]. Subtype 0, which is enriched for *EGFR* mutations and shows suppressed cell-cycle activation, resembles the Terminal Respiratory Unit (TRU) subtype, characterised by a differentiated transcriptional state and *EGFR*-driven oncogenesis. Subtype 1, with its dominant E2F and G2-M programmes, parallels the Proximal Proliferative (PP) subtype. Subtype 2, defined by interferon signalling activation, is broadly consistent with the Proximal Inflammatory (PI) subtype, although the co-occurrence of E2F upregulation in subtype 2 does not map cleanly onto classical PI biology. Importantly, state-based and trajectory-based subtypes are conceptually distinct: TRU, PP, and PI describe what a tumour looks like, whereas the MOSAIC subtypes describe how the patient got there. The parallels are therefore informative rather than expected, and their correspondence warrants formal testing in a larger dataset.

The mutation enrichment result provides an independent line of evidence for the biological coherence of the subtype structure. Because *EGFR* mutations and ALK rearrangements are mutually exclusive oncogenic events in LUAD, the observation that both are differentially distributed within the same subtype (enriched and depleted, respectively, in subtype 0) implies that the clustering captures biologically meaningful patient groupings rather than statistical artefacts.

The L2 normalisation choice and cluster geometry. Clustering was performed on L2-normalised ΔZ , which focuses on transformation *direction* rather than magnitude. The practical consequence is that two patients with very different transformation extents but qualitatively similar factor profiles are assigned to the same subtype, while magnitude is treated as an independent dimension. This choice is supported by the absence of a significant association between transformation magnitude and tumour stage, which argues against magnitude being a simple proxy for disease state. The resulting silhouette scores (≈ 0.18) are modest, reflecting a partially continuous manifold rather than discrete clusters; this is expected for a cohort of 100 patients stratified by a 13-dimensional biological signal.

Limitations. The most consequential limitation is sample size. With $n = 102$ patients, many of whom were censored before the study endpoint, the study is underpowered for survival analysis and the null log-rank result is uninformative rather than negative evidence. Biological interpretation relies exclusively on transcriptomic data at two stages: GSEA on MOFA factor loadings uses RNA weights only, and cluster differential expression uses tumour-minus-normal RNA rather than the multi-omics signal used in the MOFA fit. The proteomic and phosphoproteomic modalities therefore contribute to the latent space but

are currently absent from the interpretation layer, leaving a portion of the multi-omics information unexploited. Finally, all results are derived from a single cohort; in the absence of external validation, the reported subtypes should be treated as hypotheses rather than established molecular entities.

Future directions. Multi-cancer validation using other CPTAC cohorts (e.g. breast or colorectal) would test whether latent trajectory structure is a general property of matched multi-omics tumourigenesis or specific to LUAD. Kinase activity inference from phosphoproteomics factor loadings (e.g. using kinase-substrate databases such as PhosphoSitePlus) would bring the third omics modality into the interpretation layer and link transformation subtypes to specific regulatory programmes. Fitting a continuous manifold model on ΔZ vectors, such as diffusion pseudotime, would complement the discrete clustering by revealing whether the three subtypes represent distinct attractors or points along a shared trajectory. The stochastic nature of MOFA fitting also deserves attention: comparing factor structures across models fitted with different random seeds would clarify which factors are stable properties of the data and which are more sensitive to initialisation. Finally, the current analysis uses MAD-based preselection on three omics views. Expanding the feature selection strategy or incorporating additional views (e.g. copy-number alterations or miRNA) may better separate the transformation modes and reduce the current dependence on a single, view-agnostic ranking criterion.

5 Conclusions

The MOSAIC pipeline demonstrates that per-patient ΔZ vectors derived from a matched tumour-normal multi-omics cohort can support biologically informative patient stratification. Three reproducible transformation subtypes were identified from the CPTAC-LUAD dataset, each characterised by a distinct transcriptional reprogramming signature. Subtype 0 shows relatively suppressed cell-cycle activation and is significantly enriched for *EGFR* mutations. Subtype 1 is proliferation-dominant, with strong E2F and G2-M up-regulation. Subtype 2 is characterised by interferon-driven immune activation combined with a proliferative component. The *EGFR* enrichment in subtype 0 gains additional credibility from the simultaneous depletion of ALK fusions in the same subtype: because *EGFR* mutations and ALK rearrangements are mutually exclusive driver events, observing both associations together strengthens confidence in the subtype structure rather than attributing either result to chance alone.

The geometric separation between subtypes is modest (mean silhouette ≈ 0.18), suggesting a partially continuous biological manifold rather than fully discrete clusters. No significant overall survival difference was detected, which reflects the limited statistical power of the study ($n = 102$) rather than necessarily indicating clinical irrelevance. Replication in an independent LUAD cohort with matched normal profiling would be the most informative next step.

Code availability. The full analysis pipeline is publicly available at <https://github.com/ajanowiak/MOSAIC>.

References

- [1] Cancer Genome Atlas Research Network. Comprehensive molecular profiling of lung adenocarcinoma. *Nature*, 511:543–550, 2014. doi: 10.1038/nature13385.
- [2] Michael A Gillette, Shankha Satpathy, Song Cao, Saravana M Dhanasekaran, Suhas V Vasaikar, Karsten Krug, Francesca Petralia, Yize Li, Wen-Wei Liang, Boris Reva, et al. Proteogenomic characterization reveals therapeutic vulnerabilities in lung adenocarcinoma. *Cell*, 182(1):200–225, 2020. doi: 10.1016/j.cell.2020.06.013.
- [3] Ricard Argelaguet, Damien Arno, Danila Bredikhin, Yosef Deloro, Britta Velten, John C Marioni, and Oliver Stegle. MOFA+: a statistical framework for comprehensive integration of multi-modal single-cell data. *Genome Biology*, 21(1):111, 2020. doi: 10.1186/s13059-020-02015-1.

A Supplementary Figures

A.1 Preprocessing

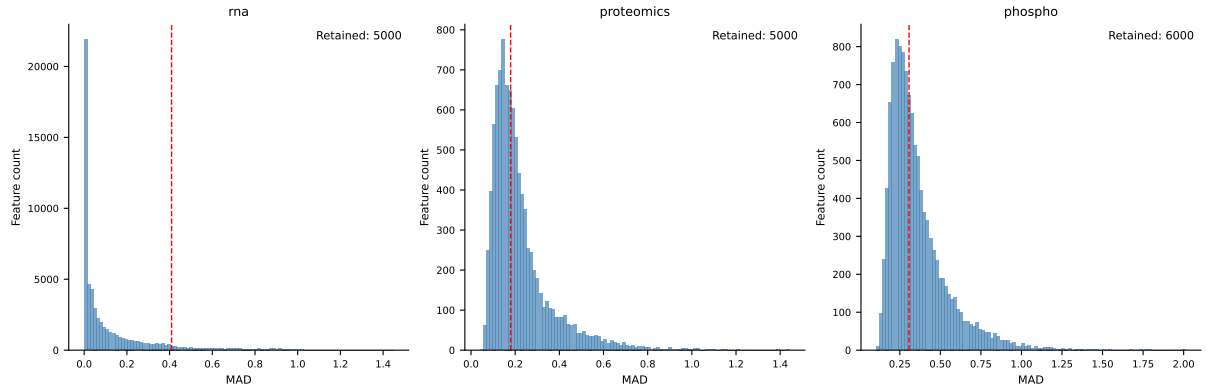


Figure A1: MAD distributions for each omics view after ingestion. The dashed red line marks the MAD of the last retained feature; all features to the right were selected. Retained counts: 5,000 transcripts, 5,000 proteins, 6,000 phosphosites.

A.2 MOFA model fitting

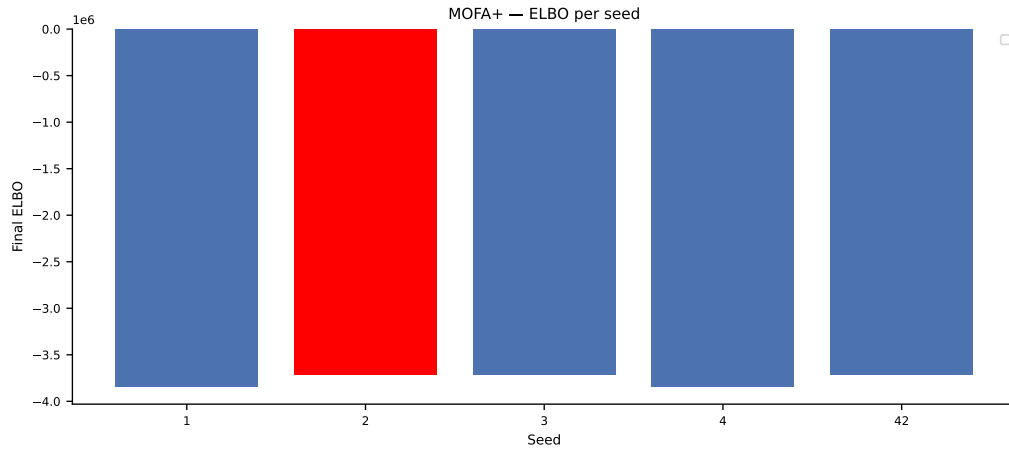


Figure A2: Final ELBO per random seed. The model with the highest ELBO (seed 2, -3.72×10^6) was selected for all downstream analyses.

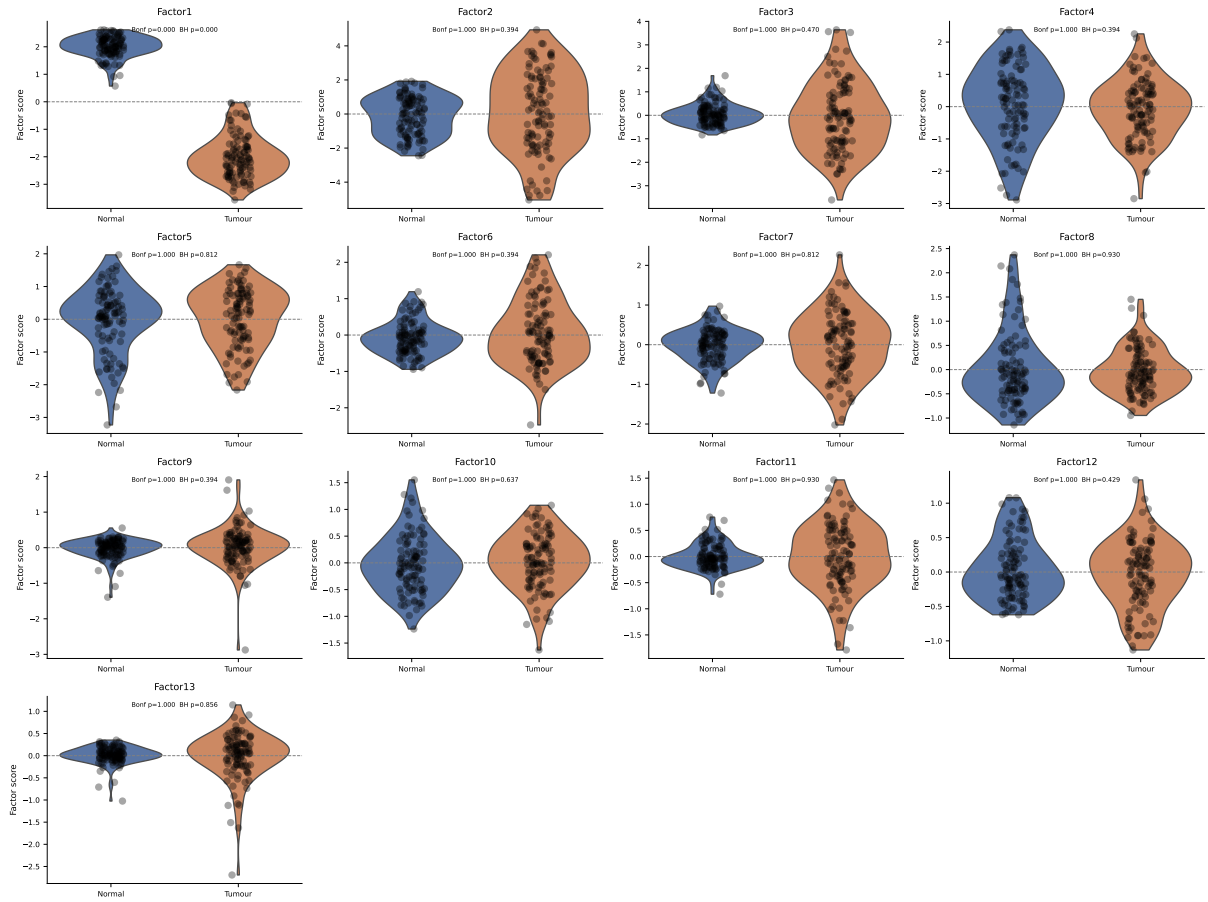


Figure A3: Distribution of MOFA+ factor scores split by tissue type (tumour vs. normal) for all 13 retained factors. Factor 1 shows complete separation; no other factor reaches significance after Bonferroni correction.

A.3 Tumourigenesis trajectories

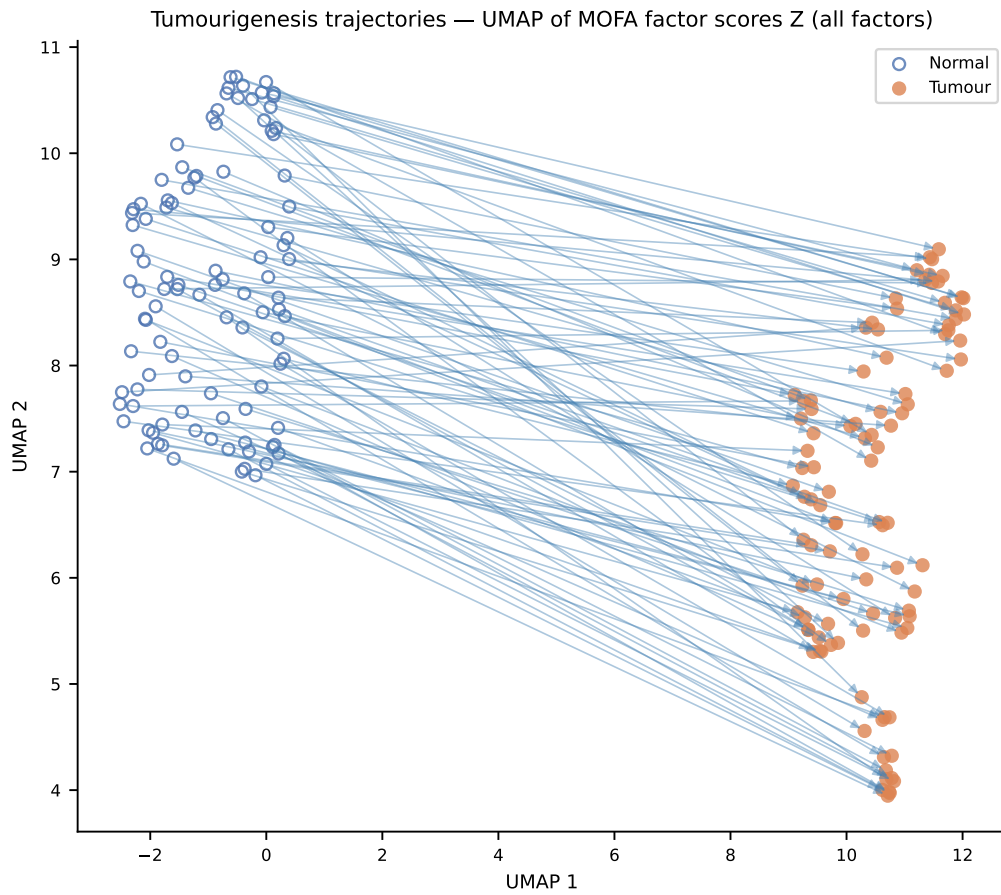


Figure A4: Per-patient tumourigenesis trajectories in the UMAP of MOFA+ factor scores (all 13 factors, all 204 samples). Each arrow connects the normal to the tumour sample of one patient.

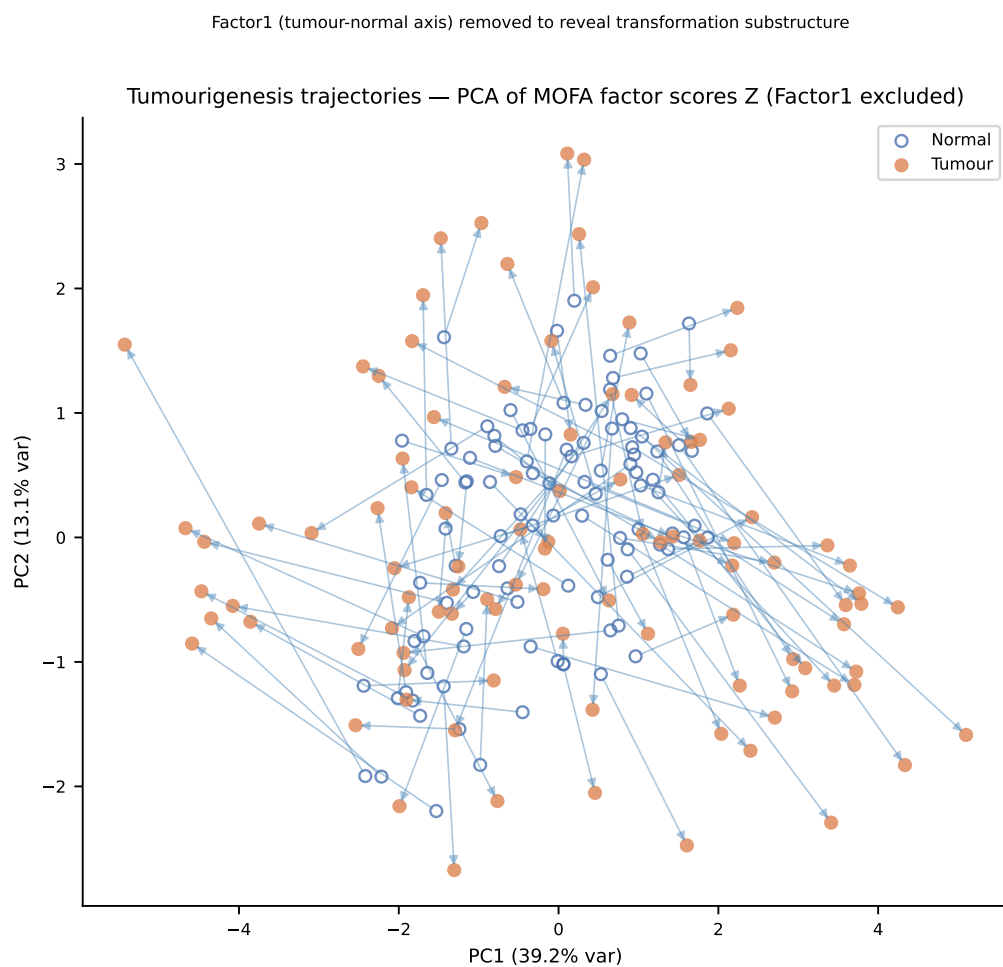


Figure A5: Trajectory arrows in PCA of factor scores with Factor 1 excluded. Projecting out the dominant tumour-normal axis reveals substructure in the remaining factors that is obscured in Figure 2.

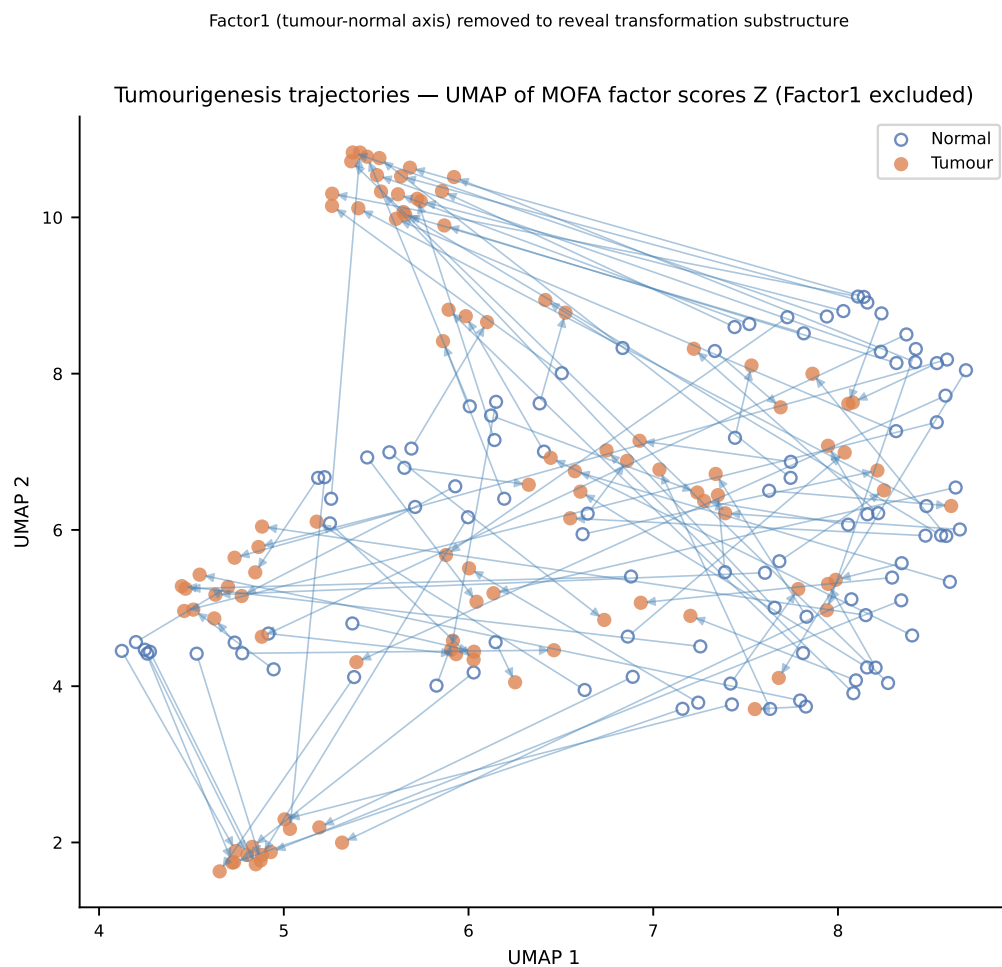


Figure A6: Trajectory arrows in UMAP of factor scores with Factor 1 excluded. Arrow directions are more heterogeneous than in the full-factor UMAP (Figure A4), confirming that Factor 1 drives the dominant shared trajectory.

A.4 ΔZ heterogeneity

Per-factor ΔZ distributions across patients (n=102)

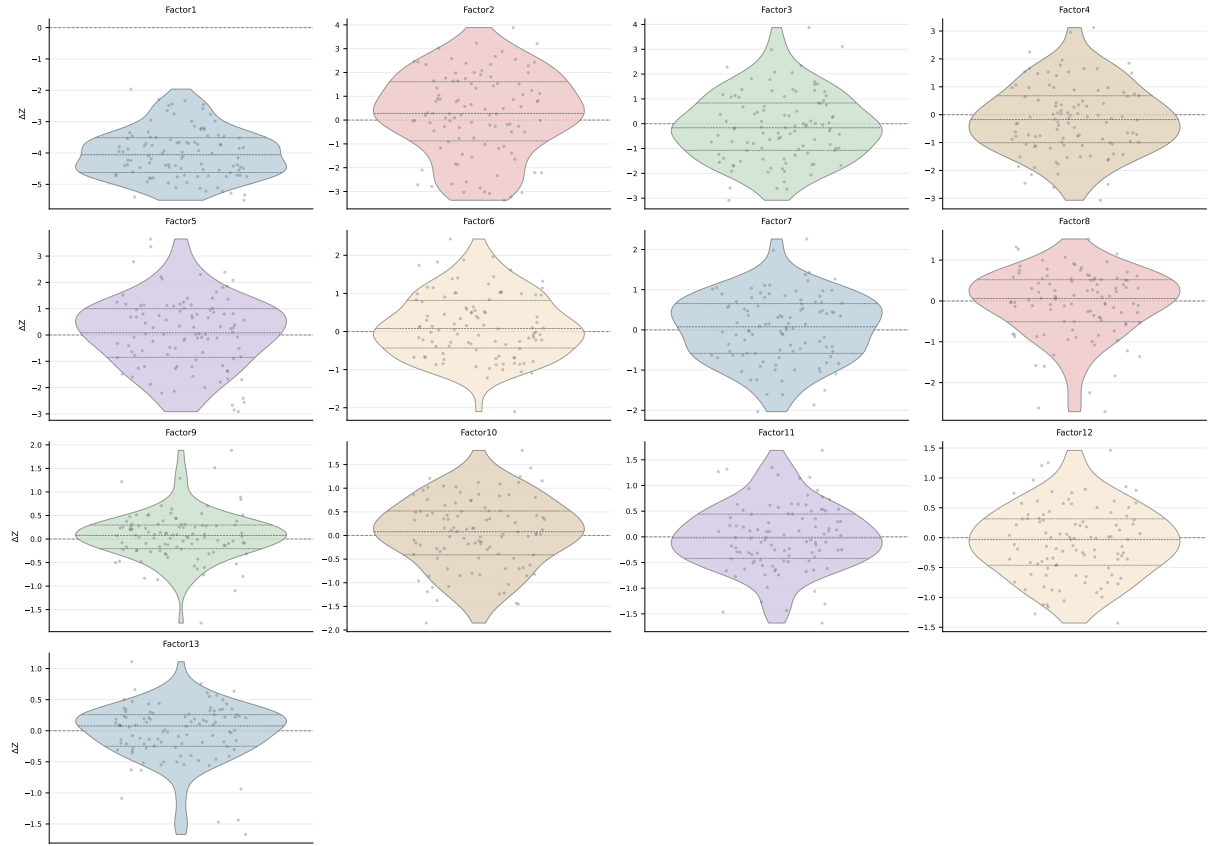


Figure A7: Per-factor ΔZ distributions across all 102 patients. Factor 1 is strongly negative in all patients (universal tumourigenesis shift); the remaining factors show patient-level variation of varying magnitude.

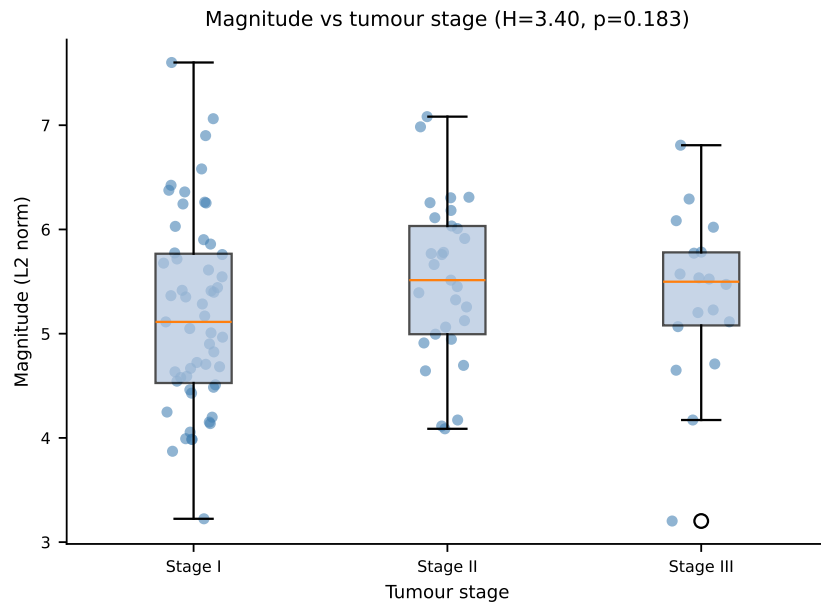


Figure A8: L2 norm of ΔZ (transformation magnitude) stratified by tumour stage. No significant difference was observed (Kruskal-Wallis $H = 3.40$, $p = 0.183$).

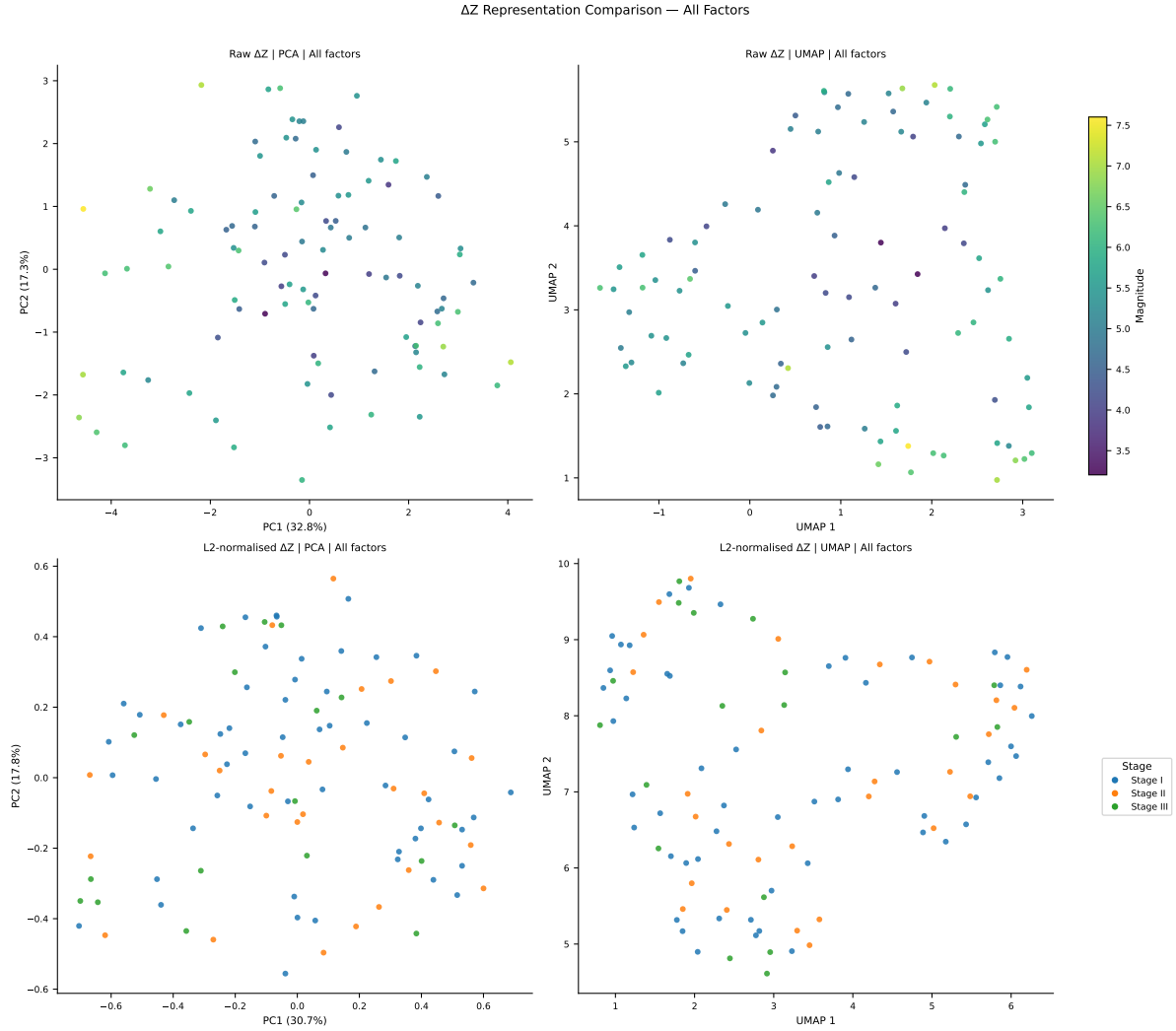


Figure A9: 2×2 comparison of ΔZ representations: raw vs. L2-normalised × PCA vs. UMAP, using all 13 factors. Points are coloured by transformation magnitude (viridis). The panel was used to select L2-normalised ΔZ as the primary clustering representation.

A.5 Clustering

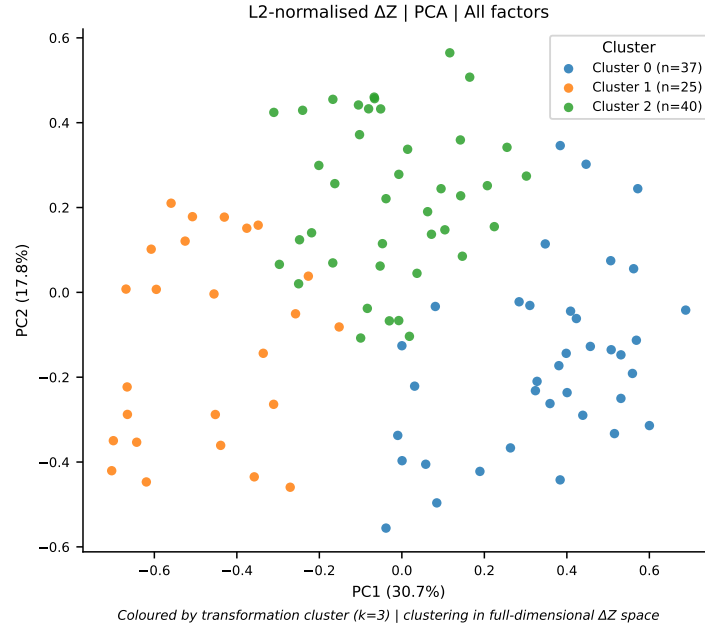


Figure A10: PCA of L2-normalised ΔZ coloured by transformation subtype. The three subtypes overlap substantially in 2D, consistent with the UMAP in Figure 4 and the low silhouette scores.

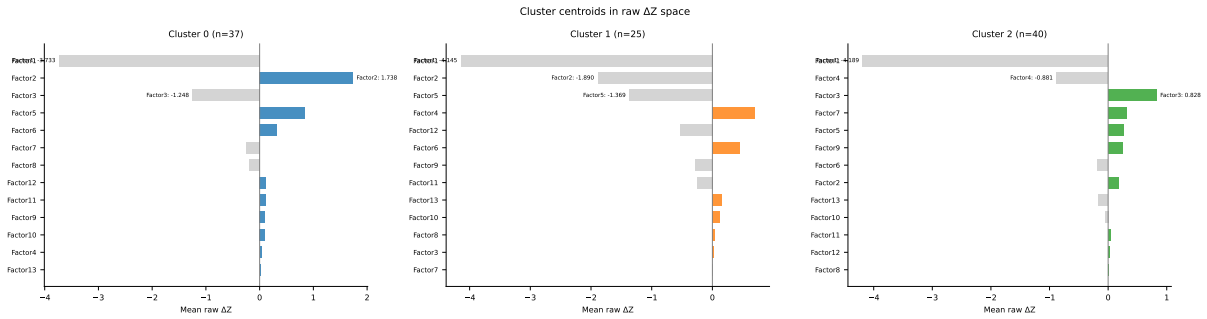


Figure A11: Factor-level centroid profiles for each transformation subtype in raw ΔZ space. All subtypes have a strongly negative Factor 1 component; differences in Factors 2–13 define the subtype-specific patterns.

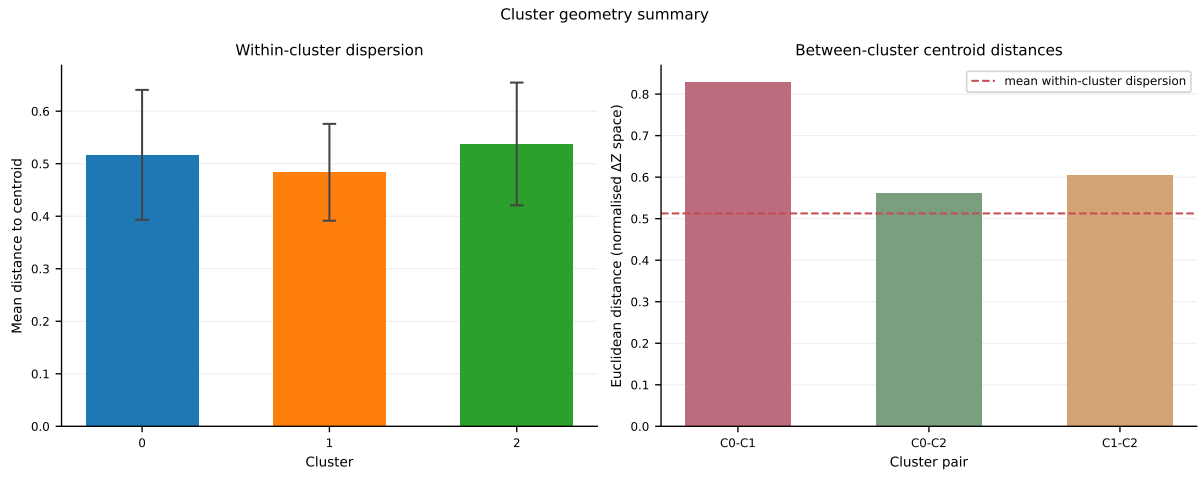


Figure A12: Within-cluster dispersion (left) and pairwise centroid distances (right) in L2-normalised ΔZ space. Centroid distances (0.56–0.83) exceed mean within-cluster radii (0.48–0.54) for all pairs, indicating that the subtypes are more separated than they are internally dispersed.

A.6 Biological interpretation

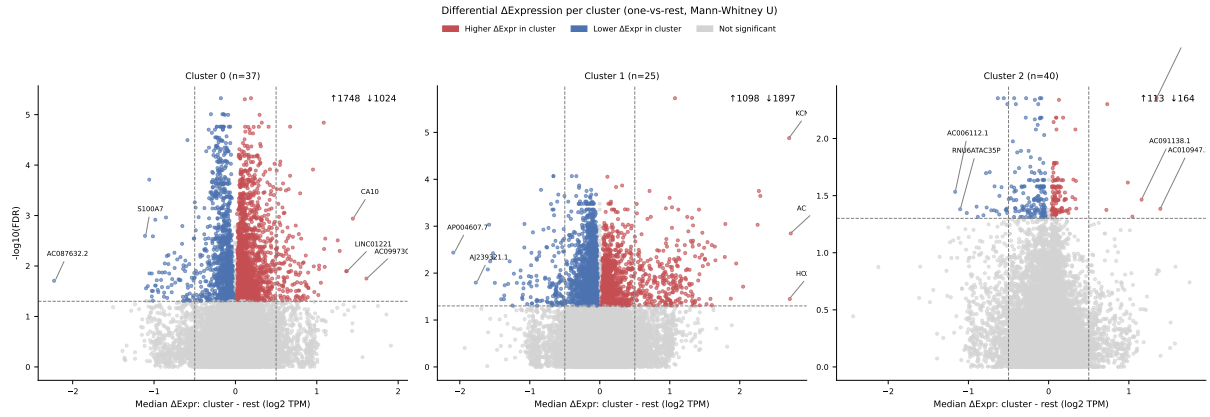


Figure A13: Volcano plots of one-vs-rest Δ Expression for each transformation subtype. Each point is a gene; x -axis shows median difference in Δ Expr between the focal subtype and all other patients; y -axis shows $-\log_{10}(\text{FDR})$. Top differentially expressed genes are labelled.

A.7 Clinical associations

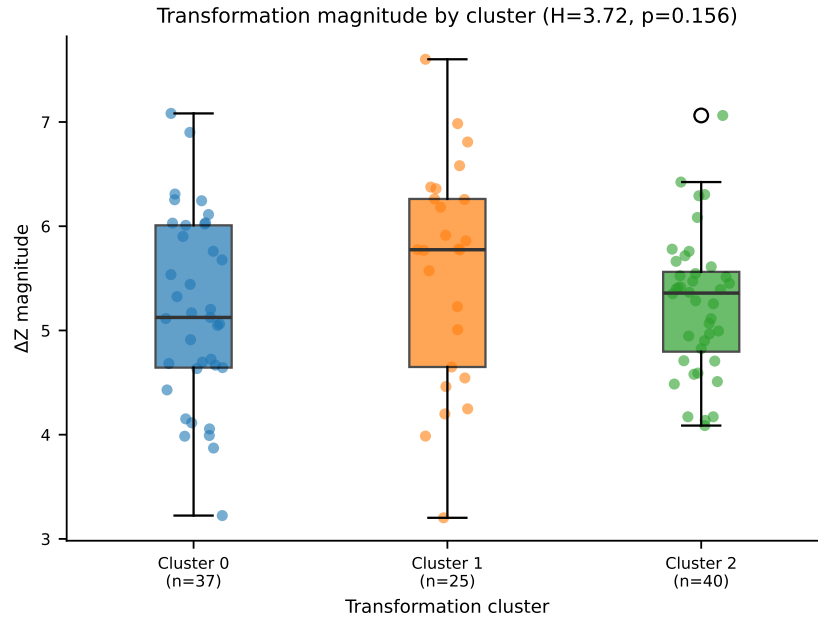


Figure A14: Transformation magnitude (L2 norm of ΔZ) per subtype. No significant difference was observed across subtypes (Kruskal-Wallis $H = 3.72$, $p = 0.156$).

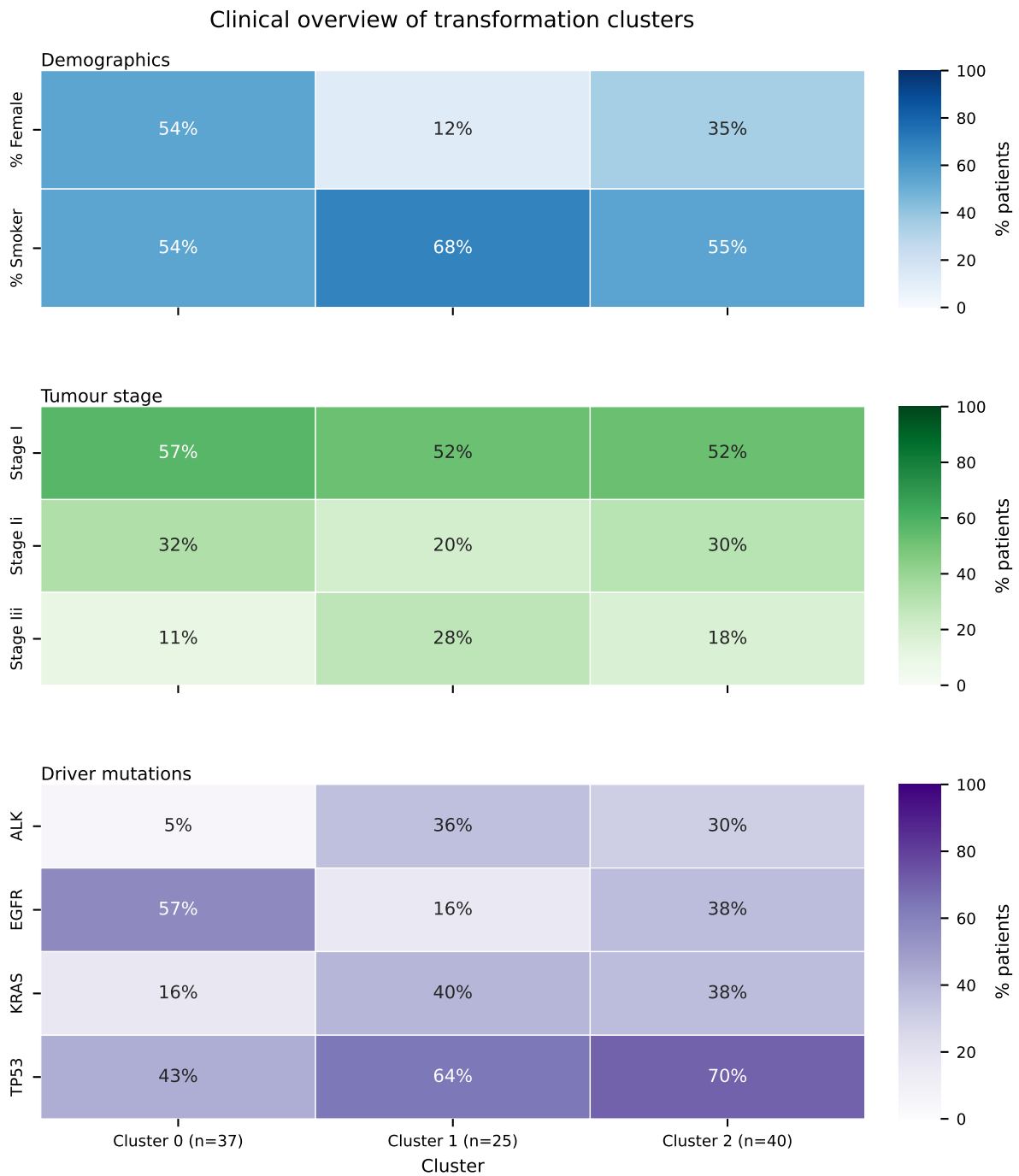


Figure A15: Descriptive clinical characteristics per transformation subtype. Three panels show demographics (sex, age, smoking), tumour stage distribution, and driver mutation prevalence. Values are percentages; median age is reported separately in Table A2.

B Supplementary Tables

B.1 Factor tissue separation statistics

Table A1: Paired Wilcoxon test for tumour-vs-normal separation per MOFA+ factor, applied to ΔZ values across all 102 patients. p -values are corrected by Benjamini-Hochberg (BH). Only Factor 1 is significant after correction.

Factor	W	Raw p	Rank-biserial r	BH-adjusted p
Factor1	0	1.8×10^{-18}	1.000	2.4×10^{-17}
Factor2	2174	0.131	0.172	0.394
Factor3	2284	0.253	0.130	0.470
Factor4	2190	0.145	0.166	0.394
Factor5	2453	0.562	0.066	0.812
Factor6	2160	0.119	0.178	0.394
Factor7	2480	0.625	0.056	0.812
Factor8	2600	0.930	0.010	0.930
Factor9	2197	0.152	0.164	0.394
Factor10	2370	0.392	0.098	0.637
Factor11	2576	0.866	0.019	0.930
Factor12	2241	0.198	0.147	0.429
Factor13	2521	0.725	0.040	0.856

B.2 Clinical characteristics per subtype

Table A2: Clinical characteristics per transformation subtype. Values are percentages except median age. Smoking: ever-smoker (current or reformed). Cluster 0 has 34 patients with available survival data (3 missing).

	Subtype 0 ($n = 37$)	Subtype 1 ($n = 25$)	Subtype 2 ($n = 40$)
Median age (years)	66	58	65.5
Female (%)	54.1	12.0	35.0
Ever-smoker (%)	54.1	68.0	55.0
Stage I (%)	56.8	52.0	52.5
Stage II (%)	32.4	20.0	30.0
Stage III (%)	10.8	28.0	17.5
<i>KRAS</i> mut. (%)	16.2	40.0	37.5
<i>EGFR</i> mut. (%)	56.8	16.0	37.5
<i>TP53</i> mut. (%)	43.2	64.0	70.0
ALK fusion (%)	5.4	36.0	30.0

B.3 Kaplan-Meier at-risk table

Table A3: Kaplan-Meier at-risk table. At-risk counts at each time point; cumulative censored and event counts are shown for each subtype. Subtype 0 has 34 patients with survival data available.

Days	Subtype 0			Subtype 1			Subtype 2		
	At risk	Censored	Events	At risk	Censored	Events	At risk	Censored	Events
0	34	0	0	25	0	0	40	0	0
250	30	3	1	23	0	2	36	3	1
500	23	9	2	9	12	4	17	19	4
750	21	10	3	6	15	4	13	21	6
1000	17	11	6	3	17	5	11	22	7
1250	10	18	6	2	18	5	7	25	8
1500	8	20	6	1	19	5	6	26	8
1750	4	23	7	1	19	5	4	28	8
2000	0	27	7	0	20	5	0	32	8